Ridership Data Collection and Exploration Report (First Round)

Introduction

The Ridership Data Collection Plan was launched to improve data quality and granularity with a focus on stop-level ridership. Following the roadmap outlined in the plan, we have completed efforts 1, 2 and 10, which involved data collection and discovery of highest-ridership agencies across the state. The objective of the efforts is to evaluate whether it is feasible to align agency-provided ridership data to a common structure, either by adopting an existing standard (such as TIDES or GTFS-Ride) or by identifying the need for a new or modified standard that agencies can use to pull ridership data on a recurring basis.

In this round, we collected and analyzed stop-level ridership datasets from 19 agencies, representing a significant share of statewide ridership. Following data collection, each agency's dataset was reviewed to understand what ridership fields were provided, the level of granularity, the temporal coverage of the data, and common data gaps and issues. To produce a combined stop-level ridership dataset (Effort 10), agency-specific transformations tailored to the structure and content of each dataset were applied. In parallel, the range of data formats and structures was examined to evaluate whether any common ridership data standards currently exist across agencies and how well existing standards could accommodate the data agencies are able to provide.

This document describes the scope, methods, and outcomes of Effort 1, 2 and 10. It summarizes the data collected, outlines processing and aggregation approaches used to produce the combined stop-level ridership dataset, documents key data limitations, and implementation challenges, which can inform potential improvements to data collection, processing workflow and standardization approaches.

The current effort is intended to be exploratory and foundational. The resulting dataset reflects the data that agencies were able to provide during this initial collection and is not intended to represent complete and finalized ridership reporting. It serves as a baseline for understanding current data availability and informing future iteration of the project.

Data Collection Overview

In the first round, ridership datasets were collected from **19** transit agencies across California. Agencies were selected based on ridership levels, with the intent of prioritizing agencies that represent highest share of statewide transit ridership. Participation was voluntary and dependent on agencies' willingness and ability to share detailed ridership data within the project timeline.

A summary of dataset granularity and temporal coverage can be found in Table 1 and Table 2.

Table 1 Summary of Dataset Geographic Granularity

Granularity	Description	Number of Datasets
Stop	Each record represents ridership at a specific stop	13
Trip-stop	Each record represents ridership at a specific stop within a specific trip	5
Transaction	Each record represents a single transaction or tap. Ridership is obtained by counting records over a given stop.	1

Table 2 Summary of Dataset Temporal Granularity and Coverage

Reporting Temporal Unit	Ridership Value Representation	Description	Number of Datasets
Daily	Native daily ridership	Ridership reported at a daily level in source data	8
Monthly	Average daily	Average daily ridership reported for monthly periods	2

Monthly	Monthly Total	Ridership reported as total counts for a month	1
Annual/Fiscal year	Annual/Fiscal year totals	Ridership reported as total counts for a year	2
Annual/Fiscal year	Average daily	Average daily ridership reported for a year	2
Custom service period (e.g. seasonal schedules)	Average daily (or time-of-day periods)	Average daily (or time-of-day periods) ridership reported for agency-defined service periods	4

The following table summarizes the number of collected datasets that are missing key stop-level attributes, highlighting common gaps in stop identifiers, names, and geographic coordinates across the data received in this round of collection.

Table 3 Summary of Missing Stop-Level Attributes

Missing Stop Attributes	**Number of Datasets Missing Attribute**
Stop ID (expect to be consistent with GTFS)	4
Stop name	2
Stop latitude and longitude	12

A list of agencies can be found in the ***ridership_dataset_metadata.csv*** with the characteristics of the received dataset. Details about this metadata table can be found in the Ridership dataset Metadata table section.

Data Processing and Integration Workflow

In addition to agency-reported information, there are data gaps and integration challenges identified during analysis. To produce a combined stop-level ridership dataset from heterogeneous agency-provided inputs, data processing was conducted in three sequential steps: initial ingestion and standardization, dimension enrichment and final aggregation.

STEP 1 – INITIAL INGESTION AND STANDARDIZATION

The first processing step focused on importing raw agency-provided datasets and applying minimal transformation necessary to support downstream analysis. The goal of this step was to standardize heterogeneous inputs into a consistent long-format dataset with a shared schema, while preserving raw data values wherever possible. This ensures that datasets from all agencies could be combined into a single, unified dataset for downstream processing and analysis.

Processing at this stage addressed practical differences across agency submissions and was tailored to each dataset. Route and direction level fields were retained where provided, and agency-reported stop identifiers were preserved. Table 4 summarizes the types of transformations applied at this stage.

Table 4 Summary of Initial Data Ingest and Standardization

Processing Category	Description	Applied When	Residual Caveats
Raw data ingestion	Import agency-provided files in original formats and combine multiple files provided by a single agency	All datasets	For the files that are too large or don't have consistent and clear columns, select a subset of records to include 2025 data where available
Header handling	Identify and apply correct header rows in the raw files	Shifted or multi-level headers	
Column inference	Infer column names based on observed values	Missing headers	Inferred column names can be wrong

Granularity alignment	Aggregate trip-stop, stop-event or transaction level data to stop level (differentiate route and direction if data come with these information)	Trip-stop, stop event, or transaction level inputs	
Temporal normalization	Aggregate ridership/boardings/alightings to average daily values	Non-daily inputs	
Raw Stop Coordinate Aggregation	If multiple coordinates exist for a single stop in the raw dataset, the maximum of latitude and longitude are selected to ensure a single coordinate pair per stop. Coordinates may later be overwritten by GTFS data during the next step if a valid stop ID/name match is found in GTFS.	Raw datasets contain multiple coordinate values per stop	If the stop cannot be matched to GTFS, the coordinates from raw data are retained.
Identifier extraction	Extract stop or route identifiers embedded in names	Identifiers embedded in text	Agency-provided identifiers may not be consistent with identifiers in GTFS
Reshaping	Convert wide-format data to long format	Wide inputs	
Column standardization	Rename columns and standardize data types	All datasets	

Day type derivation	Derive service day type (weekday, weekend or holiday) from data columns	Where applicable	
Date range derivation	Derive start and end dates of the ridership reporting period represented by each record	All datasets	

STEP 2 – DIMENSION ENRICHMENT

The second processing step addressed missing stop-level attributes in the raw agency submissions by enriching records with stop IDs, stop names, and stop latitude/longitude sourced from GTFS data. Enrichment was applied only where these fields were absent in the raw datasets, using direct joins on agency-provided stop IDs when they were present and aligned with GTFS, and falling back to normalized stop name matching when stop IDs were not provided or inconsistent.

Enrichment was performed using the most recent available GTFS data at the time of processing and didn't account for historical changes in stop definitions over time (e.g., stop ID, name or coordinates). As a result, while this approach was applied consistently to datasets with missing stop attributes, some records could not be matched, and others may reflect less accurate matches when ridership data corresponds to earlier service periods. These limitations are summarized in the following table that describes detailed processing steps and any cases that were not resolved.

Table 5 Summary of Dimension Enrichment Method

Processing Category	**Description**	**Matching/Joining Key**	**Residual Caveat**
Stop attribute mapping with stop ID present	Map stop name, latitude and longitude from GTFS to where they are missing in agency-provided datasets.	Stop ID/Code	When an agency provided stop IDs that are not consistent with GTFS data, it is not possible to map in other attributes from GTFS using stop ID. (Agency may provide stop code as stop ID)

	For latitude and longitude that comes from vehicle location, take GTFS latitude and longitude.		Stop attributes were mapped using the most recent GTFS data and do not account for historical stop changes in GTFS, which may result in missing or less accurate matches.
Stop attribute mapping with stop ID missing	Map stop ID, latitude and longitude from GTFS to where they are missing in agency-provided datasets. For latitude and longitude that comes from vehicle location, take GTFS latitude and longitude.	Stop name	Stop attributes were mapped using the most recent GTFS data and do not account for historical stop changes in GTFS, which may result in missing or less accurate matches.

STEP 3 – STOP-LEVEL AGGREGATION AND AVERAGE DAILY RIDERSHIP DERIVATION

After datasets from all agencies were combined and stop-level attributes enrichment was completed, ridership records are aggregated to the stop level. When multiple records are associated with the same stop, such as when the stop is served by multiple routes or directions, ridership values are summed at the stop level.

Average daily ridership was then assigned to each stop-level record. When daily-level ridership was available, it was derived by averaging daily observations within the corresponding time period. In other cases, it may be calculated from a reported total over the period or retianed from source-reported average, as indicated in the ***avg_daily_method*** column in the output dataset.

For records based on source-reported averages or totals over a period, multiple reporting periods may be available for the same stop, for example, multiple months, quarters or fiscal years. In these cases, only the most recent available period was retained.

The above implementation makes sure that each stop has a single record for a given organization/service, time period, and day type combination. A unique key was then generated for

each service – stop (stop ID, GTFS stop id and stop name) – day type – period combination and it should be unique for each record in the output combined ridership dataset.

Caveat:

Average daily ridership values were derived using different source time spans depending on data availability. As a result, the represented time span varies across stops and agencies, and cross-agency comparisons should be interpreted accordingly.

Output Dataset

Based on the datasets we collected this round, we produced two primary output datasets to support understanding of the data received and initial exploration toward a combined stop-level ridership publication.

RIDERSHIP DATASET METADATA TABLE

To capture the differences across agency-provided datasets, a metadata table was created as part of the data output at current stage. This table serves as the primary reference for understanding the format, structure, coverage, data collection methods and known data gaps provided by agencies.

For each agency and service, the metadata table includes:

Table 6 Data Description of Metadata Table (1)

Column Name	Description
dataset_id	Unique identifier for each agency-provided dataset
dataset_name	Name of the initial processed (first step processing) dataset derived from the raw agency submission.
organization_name	Name of the transit agency that provided the dataset
service_name	Name of the service associated with the dataset

start_date_collected	Earliest date of ridership data included
end_date_collected	Latest date of ridership data included
reporting_unit	The temporal unit at which ridership values are expressed within the dataset, e.g., values reported for a day, month, year, or custom period
ridership_measure	Describes how ridership values should be interpreted for each reporting unit, e.g., daily values, totals or average daily values
geographic_grain	The spatial or operational unit represented by each record, e.g., stop, trip-stop or transaction
notes	Agency-reported data collection methods, known data gaps, or other contextual notes relevant to interpreting the dataset
gtfs_url	The URL for the GTFS dataset used for matching and obtaining stop attributes for ridership datasets with missing fields.
gtfs_snapshot_date	The date that the combination of organization, service and GTFS feed was present in the data warehouse.
gtfs_dataset_name	The GTFS dataset name that used for matching and obtaining stop attributes for ridership datasets with missing fields.
stop_corrds_from_gtfs	Indicator for whether the stop name in the output ridership table is from GTFS data.
stop_coords_from_gtfs	Indicator for whether the stop latitude and longitude in the output ridership table are from GTFS data.

For selected ridership-related fields, the metadata table includes boolean indicator columns following the naming convention *<field>_exists*. These indicators denote whether the corresponding field is present in the agency-provided dataset. Indicator columns are provided for the following fields:

Table 7 Data Description of Metadata Table (2)

Column Name	Description
route_id_exists	Indicator for whether the agency-provided dataset includes a route ID field
route_name_exists	Indicator for whether the agency-provided dataset includes a route name field
direction_exists	Indicator for whether the agency-provided dataset includes a direction field
stop_id_exists	Indicator for whether the agency-provided dataset includes a stop ID field
stop_name_exists	Indicator for whether the agency-provided dataset includes a stop Name field
stop_lat_exists	Indicator for whether the agency-provided dataset includes a stop latitude field
stop_lon_exists	Indicator for whether the agency-provided dataset includes a stop longitude field
boardings_exists	Indicator for whether the agency-provided dataset includes a boardings field
alightings_exists	Indicator for whether the agency-provided dataset includes an alightings field
ridership_exists	Indicator for whether the agency-provided dataset includes a ridership field

This metadata table is intended to be used alongside the combined stop-level ridership dataset. Analysts and downstream users can reference this table to identify which data elements are

available for a given agency/service, understand shared patterns and differences in raw data and inform filtering, aggregation and comparison decisions in analysis.

COMBINED STOP-LEVEL RIDERSHIP TABLE

After applying all the processing steps to the raw datasets, average daily ridership for consistent stop-level granularity across agencies was combined to one dataset. The schema is as follows:

Table 8 Data Description of Combined Ridership Table

Column Name	Description
record_id	Unique identifier for each stop, service day type and reporting period combination
dataset_id	Identifier for the source agency-provided dataset associated with the record
organization_name	Name of the transit agency that provided the data
service_name	Name of the service associated with the dataset
stop_id	GTFS stop id where available
stop_name	Stop name (GTFS-mapped when agency didn't provide)
stop_lat	Stop latitude (GTFS-mapped when agency didn't provide or agency-provided data have more than one set of lat/lon for one stop)
stop_lon	Stop longitude (GTFS-mapped when agency didn't provide or agency-provided data have more than one set of lat/lon for one stop)
avg_daily_boardings	Average daily boardings for the stop and reporting period. The method used to produce the average daily value is documented in avg_daily_method column.

avg_daily_alightings	Average daily alightings for the stop and reporting period. The method used to produce the average daily value is documented in avg_daily_method column.
avg_daily_ridership	Average daily total ridership for the stop and reporting period. The method used to produce the average daily value is documented in avg_daily_method column.
day_type	Service day type associated with the ridership data (e.g., weekday, weekend, holiday, or all)
avg_daily_method	Indicates the method used to produce the average daily value. Possible values: - avg_from_daily: calculated from daily-level ridership records in the corresponding period - avg_from_total: calculated by dividing a reported total ridership by the number of days in the corresponding period - reported_avg: average daily ridership was provided directly in the source data
start_date	Start date of the time period rerpesented by the record
end_date	End date of the time period represented by the record

The metadata table provides essential context for understanding differences across agency datasets, while the combine stop-level ridership dataset provides the ridership values for analysis. Two datasets share a common *dataset_id* column.

Future Improvements

Section 3 documents data findings and caveats after initial data processing and integration. These findings highlighted several opportunities to improve future iterations of data collection and integration effort.

DATA COLLECTION

The first round of data collection demonstrated that agencies vary widely how stop level ridership data are structured, aggregated and documented. While this diversity reflects different operational needs and technical capacity, it also introduced complexity in downstream data processing and integration.

Future iterations could benefit from clearer guidance around requested data formats and schema elements, even if full standardization is not immediately feasible. In particular, to reduce ambiguity and the need for agency-specific preprocessing, we could explicitly ask agencies to provide the following information:

- Fields
    - Route
    - Direction (optional)
    - Stop ID (ideally GTFS stop ID)
    - Stop name
    - Boardings
    - Alightings
    - Ridership (required if no boardings and alightings available)
- Column names. Ideally standardize column names. Column naming conventions and descriptions would be helpful. For example, a column named stop ID may be different from GTFS stop ID.
- Clear description of the provided columns. Especially when only ridership is provided without boardings and alightings, a clear definition of "ridership" is expected.
- Description of granularity and any context/caveat of data
    - Geographic granularity. Some agencies provided fine-grained data (e.g. trip-stop level or transaction data). Where feasible, future iterations could encourage agencies to

share stop-level aggregates. Any context about how the aggregates are calculated would be helpful.

- Temporal granularity. We target daily or average daily ridership. When ridership values represent aggregates over a reporting period, documentation on how the aggregated values are calculated can improve interpretability in future iterations.
- Reporting period. Most recent available data at the time of collection is expected.

While this effort focused on partnering with agencies willing and able to share detailed stop ridership data, future rounds may also consider how emerging or existing standards (e.g. GTFS-Ride and TIDES) could be leveraged to support more consistent ridership data collection.

DATA PROCESSING AND INTEGRATION

From a data processing perspective, this initial round prioritized exploratory analysis of each agency-provided dataset to understand its structure, formats, granularity and data gaps. To accommodate the wide variability in source datasets, data processing was intentionally flexible and tailored to individual agency submissions. While this approach was necessary for initial integration, it also revealed opportunities to improve the consistency, robustness and efficiency of data processing pipeline in future iterations.

One key consideration is the use of GTFS as a reference for stop level enrichment. In the current stage, the most recent available GTFS at the time of processing was used as a sanity check to fill missing stop IDs, names and coordinates. This approach does not account for changes in stop data over time. A more precise strategy for future iterations would involve identifying and using the GTFS feed that was active during the corresponding ridership reporting period, which would improve temporal alignment and spatial accuracy in stop level enrichment.

Additionally, as more agencies participate and as preferred data patterns become clearer, there will be opportunities to streamline processing by supporting partial/full standardization such as defining a common set of core fields and limit accepted input format and aggregation patterns. This would reduce the need for agency-specific processing and automate the procedure as much as possible.